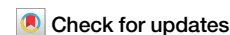


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Cracking in polymer substrates for flexible electronic devices and its mitigation



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Mechanical reliability plays a critical role in determining the durability of flexible electronic devices because of the significant mechanical stresses they experience during manufacturing and operation. Many such devices are built on sheets comprising stiff transparent-conducting oxide (TCO) electrode films on compliant polymer substrates, and it is generally assumed that the high-toughness polymer substrates do not crack. Contrary to this assumption, here we show extensive cracking in the polymer substrates during bending of a variety of TCO/polymer sheets, and a device example — flexible perovskite solar cells. Such substrate cracking, which compromises the overall mechanical integrity of the entire device, is driven by the amplified stress-intensity factor caused by the elastic mismatch at the film/substrate interface. To mitigate this substrate cracking, an interlayer-engineering approach is designed and experimentally demonstrated. This approach is potentially applicable to myriad flexible electronic devices, with stiff films on compliant substrates, for improving their durability and reliability.

There is growing interest in flexible electronic devices used in a wide range of applications such as^{1,2}: energy harvesting and storage; displays and lighting; sensors and actuators; health-monitors and wearables; *etc.* By some estimates the global market for flexible electronics is expected to exceed 75 billion US dollars by 2032³. The foundation of these devices is the flexible substrate, which is typically made of polymers such as⁴: polyethylene naphthalate (PEN), polyethylene terephthalate (PET), polyimide (PI), and polydimethylsiloxane (PDMS). For many applications, such as organic photovoltaics (OPV)⁵, perovskite solar cells (PSCs)⁶, displays/lighting⁷, touchscreens⁸, and electrochromics⁹, the polymer substrate is coated with transparent-conducting oxide (TCO) thin film, which is typically made of indium-tin oxide (ITO), indium-zinc oxide (IZO), or aluminum-zinc oxide (AZO)¹⁰.

Since the mechanical reliability of flexible electronic devices built on these TCO/polymer sheets is so important, several mechanical-testing methods have been developed^{11,12}. Among them, the bending test is the most commonly used to study the mechanical behavior and failure mechanisms in flexible electronic devices¹³. This test involves bending the device around a cylindrical mandrel of radius, r , subjecting the entire active part of the device to uniaxial tension (Fig. 1a)¹³. Since the polymer substrates' toughness is several orders of magnitude higher than that of the ceramic TCO thin film, the substrates are generally not expected to crack in bending. For example,

the PEN has a specific effective work of fracture or SEWF (w_e)¹⁴ of $\sim 55,000 \text{ J}\cdot\text{m}^{-2}$ ¹⁵, whereas the equivalent toughness, G_C , of ITO is $\sim 5 \text{ J}\cdot\text{m}^{-2}$ ¹⁶.

Results

Bending tests of TCO/polymer sheets

Here, commercially available ITO/PEN sheets, which are commonly used in flexible electronic devices, were subjected to bending tests. Figure 1b is a top-view optical micrograph of the ITO/PEN sheet in the bent state ($r = 7 \text{ mm}$), where the externally applied uniaxial tensile strain (ϵ_A) at the convex surface and near-surface region is 0.0089, as calculated using the equation $\epsilon_A = t_S/2r$ (t_S is the substrate thickness; $\sim 125 \mu\text{m}$ for PEN)^{13,17}. (Supplementary Fig. 1 plots the calculated applied uniaxial tensile universal strain (ϵ_A), as a function of r , and the corresponding stress (σ_A) within the layers studied here. For analysis of multiple layers, see Supplementary Note 1¹³. Any tensile or compressive internal residual stresses within the individual layers would augment or diminish, respectively, the overall stresses in the layers in the bent state. Supplementary Table 1 lists the thicknesses of the different layers used in this study and their relevant static mechanical properties^{15,18–26}). Under this applied stress, typical array of 'channel' cracks running parallel to the bending axis within the brittle ITO film are observed in Fig. 1b. Such observations have been reported in the flexible electronics

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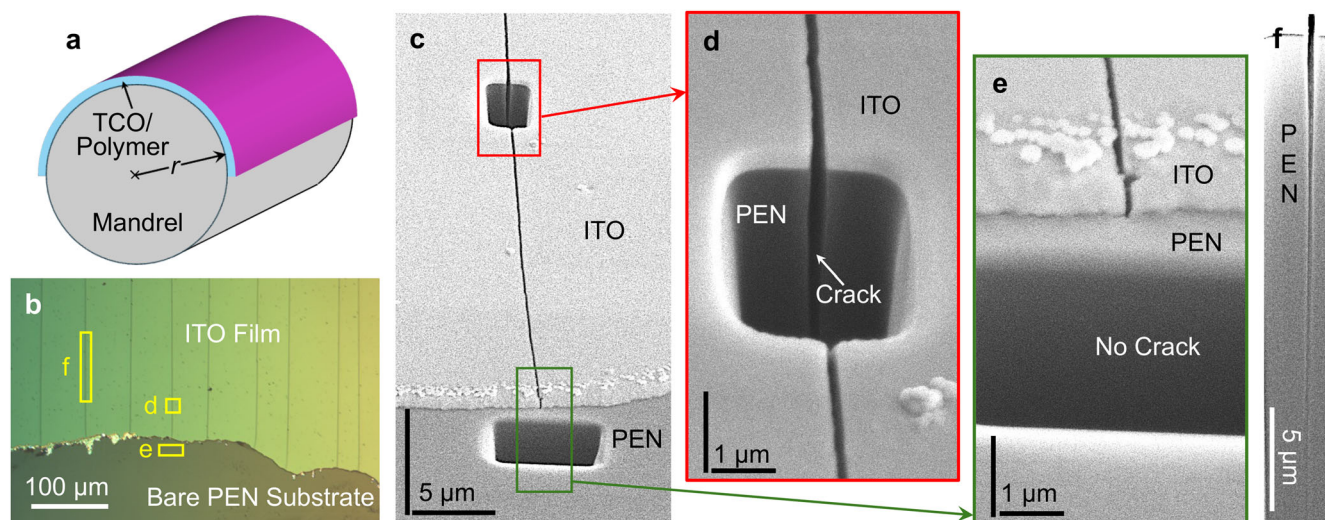
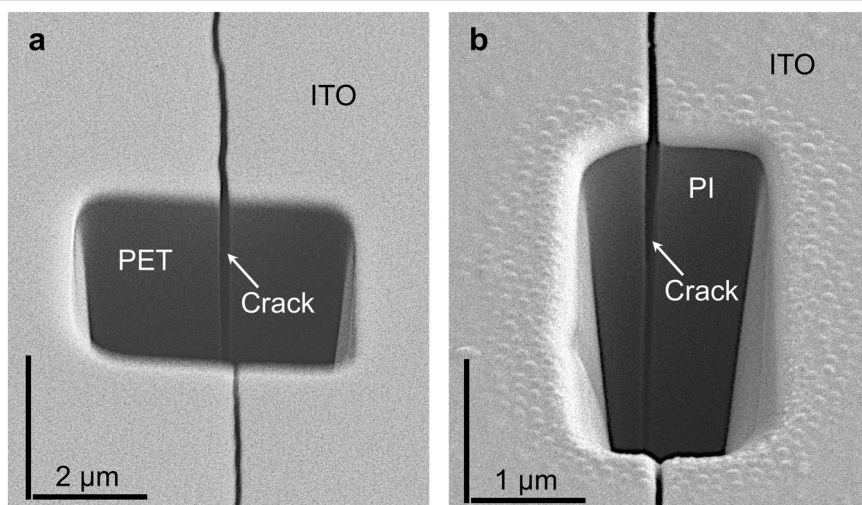


Fig. 1 | Bending tests of ITO/PEN sheets. **a** Schematic illustration (not to scale) of the bending test of ITO/PEN sheets around a cylindrical mandrel of radius, r . **b** Top-view optical microscope image of an ITO/PEN sheet bent to $r = 7$ mm ($\epsilon_A = 0.0089$), in the bent state (strained), showing ‘channel’ cracks in the ITO film and no cracks in the bare PEN substrate (bottom region). **c** SEM image of the two FIB-cuts. (FIB-cutting was performed while in the bent state from regions similar to those indicated by the yellow rectangles in **(b)**). **d, e** Corresponding higher-magnification SEM

images of each of the FIB-cuts showing, in cross-sections, cracking of both ITO film and PEN substrate (**d**) and no cracking in the bare PEN substrate (**e**). **f** Cross-sectional SEM image of a different FIB-cut showing deep cracking in the PEN substrate under ITO film. (All FIB-cut specimens were observed at a 52° forward tilt angle, hence vertical micron bars (representing depth) are longer than the horizontal ones, as indicated in **(c–f)**).

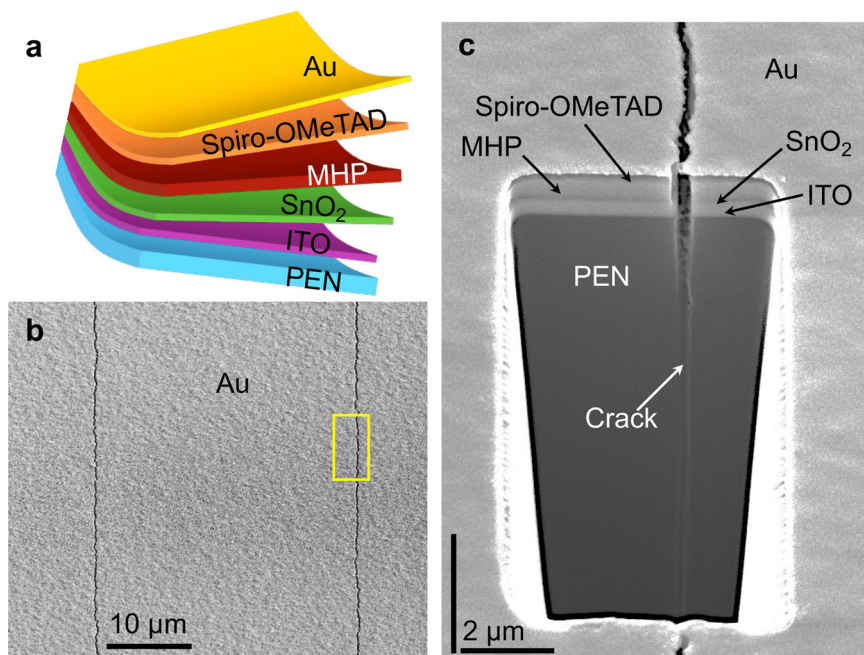
Fig. 2 | Bending tests of ITO/PEN and ITO/PI sheets. **a, b** SEM images of FIB-cut cross-sections of ‘channel’ cracks in bent ITO/PET to $r = 7$ mm ($\epsilon_A = 0.0089$) (**a**) and ITO/PI to $r = 5$ mm ($\epsilon_A = 0.0080$) (**b**) in the bent state (strained), showing cracking of both ITO films and substrates. (FIB-cutting was performed while in the bent state. All FIB-cut specimens were observed at a 52° forward tilt angle, hence vertical micron bar (representing depth) is longer than the horizontal one, as indicated).



literature on numerous occasions (see e.g. ref. 16). However, there have been no reports of substrate cracking in TCO/polymer sheets in the literature. This may be because, as mentioned earlier, it is generally assumed that the polymer substrate does not crack due to its extremely high toughness and substantial thickness relative to the TCO thin film (e.g. ITO thickness is ~ 300 nm, compared to PEN substrate thickness of $\sim 125 \mu\text{m}$)¹⁶. This purported assumption is supported by the observations in Fig. 1b where the ‘channel’ cracks in the ITO film do not appear to extend into the bare PEN substrate part without the ITO film (bottom region of Fig. 1b). However, this assumption is challenged by the scanning electron microscopy (SEM) observations of cross-sections created by focused ion beam (FIB) in Fig. 1c, d in the bent state (strained), which show cracking in the PEN substrate under the ITO film. It appears that the ‘channel’ cracks penetrate deep into the PEN substrate underneath. The cross-section SEM images in Fig. 1c, e confirm that the bare PEN substrate is uncracked. To further investigate the depth of the cracks in the PEN substrate, a longer ‘trench’ along the

crack was FIB-cut, revealing deep cracks $> 25 \mu\text{m}$ (Fig. 1f), which is $\sim 20\%$ of the total substrate thickness. To confirm that the substrate cracking was not due to the FIB-cutting process itself, part of the ITO in a previously bent ($r = 7$ mm) ITO/PEN sheet was removed by etching. Supplementary Fig. 2a, b present top-view SEM images of that specimen in the bent state, which clearly shows substrate cracking under the etched-away ITO. This type of extensive substrate cracking is observed consistently in many ‘channel’ cracks within the same ITO/PEN specimen, and also in different ITO/PEN specimens. Similar substrate-cracking behavior is observed in ITO(~ 300 nm)/PET($\sim 125 \mu\text{m}$) (Fig. 2a) and ITO(~ 200 nm)/PI($\sim 80 \mu\text{m}$) (Fig. 2b) sheets, which are also commonly used in flexible electronic devices. Furthermore, substrate cracking is observed in multilayer devices built upon ITO/PEN sheets, e.g. flexible PSCs (Fig. 3), a promising new low-cost, lightweight PV technology^{6,27,28}. It is important to note that no evidence for interfacial delamination was observed in Figs. 1d, 2a, b, and 3c. The key to the above substrate-cracking observations in the cross-sections, which have

Fig. 3 | Substrate cracking in bent flexible PSCs.
a Schematic illustration (not to scale) of the different layers in the flexible PSCs (exploded view). Here SnO_2 and Spiro-OMeTAD serve as the electron-transport and hole-transport, respectively, layers. Metal halide perovskite (MHP) serves as the light absorber material, and Au is the top electrode.
b, c SEM images of a working (19.7% efficiency) flexible PSC bent to $r = 7$ mm, in the bent state (strained), showing top-surface view of ‘channel’ cracks (**b**), and FIB-cut cross-section from an area indicated by the yellow rectangle in (**b**) of a ‘channel’ crack revealing cracking in all the layers in the flexible PSC and the PEN substrate (**c**). (FIB-cutting was performed while in the bent state from a region similar to that indicated by the yellow rectangle in (**b**). All FIB-cut specimens were observed at a 52° forward tilt angle, hence vertical micron bar (representing depth) is longer than the horizontal one, as indicated in (**c**).



eluded past studies, is that we performed FIB-cutting and SEM imaging in situ in the bent state.

This demonstration of substrate cracking in TCO/polymer sheets is significant because it raises serious questions about the prevailing notion that high-toughness polymer substrates are immune to cracking. The current design, manufacturing, and operation of flexible electronic devices are based on this notion. Since the substrate serves as the foundational ‘bedrock’ of any flexible device, substrate cracking seriously undermines the mechanical integrity and reliability of the entire device, making it susceptible to cyclic fatigue and other time-dependent failure mechanisms, such as creep and environment-assisted cracking or degradation.

Mechanics of substrate cracking

There have been a few studies reporting cracking in compliant, tough polymer substrates, such as PDMS, PET, and PI underneath either stiff, ductile metal films (Au/PDMS²⁹, Al/PDMS³⁰) or stiff, brittle ceramic films (SiN_x/PET , SiN_x/PI ³¹). (Note that while Kim et al.³¹ show visual evidence for substrate cracking in cross-sections of SiN_x/PET , those cracks formed only after holding the specimens in tension for 5 days under ambient conditions). Cracking has also been reported in compliant, tough brass substrates underneath stiff, brittle TiN films³². However, in all these cases the substrate cracking is modest, and to the best of our knowledge there have been no published reports of substrate cracking in TCO/polymer sheets and related flexible electronic devices.

To understand the genesis of the substrate-cracking phenomenon, we look back at some previous mechanics studies where cracking of compliant substrates underneath stiff films has been predicted and modeled^{33–37}. For example, analytical modeling by Gecit³⁴ indicates that when a putative crack of depth, c , in a stiff film of thickness, h , (Al or steel) under uniaxial tension approaches the interface with a compliant substrate (epoxy or Al), the crack-driving stress intensity factor, K_h , increases dramatically relative to reference K_o for no elastic mismatch between the film and the substrate. This is the result of concentrated shear stresses induced in the substrate at the crack-interface intersection due to the elastic mismatch³⁵, and it is depicted schematically in Supplementary Fig. 3a. It is implied that the substrate is expected to crack, but experimental evidence for the same is not provided in those papers^{33–37}. By corollary, when the elastic mismatch is reversed, *i.e.* compliant film (Al) on stiff substrate (steel), the K_h diminishes.

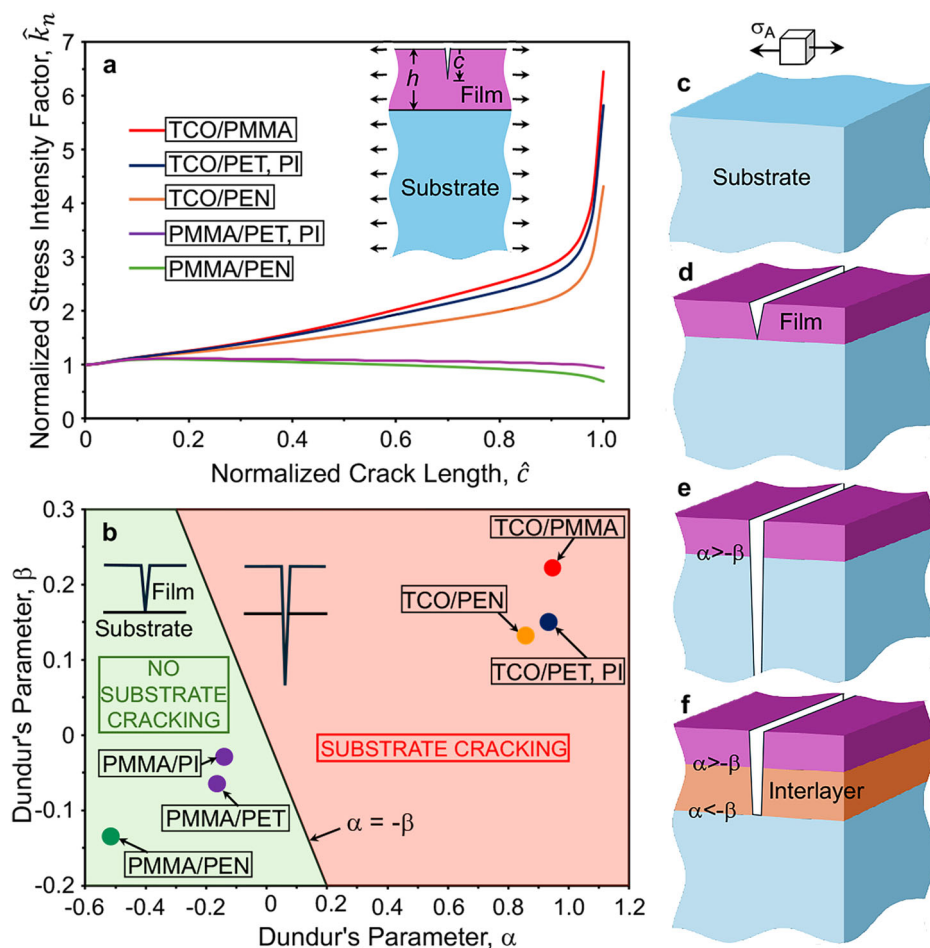
Here it is instructive to define the elastic mismatch between the film (F) and the substrate (S) in terms of the Dundur’s parameters, α and β , which incorporate not only their respective Young’s moduli (E_F , E_S) but also shear moduli (μ_F , μ_S) and Poisson’s ratios (ν_F , ν_S)³⁸:

$$\alpha = \frac{\bar{E}_F - \bar{E}_S}{\bar{E}_F + \bar{E}_S} \quad (1)$$

$$\beta = \frac{1}{2} \frac{\mu_F(1 - 2\nu_S) - \mu_S(1 - 2\nu_F)}{\mu_F(1 - \nu_S) + \mu_S(1 - \nu_F)} \quad (2)$$

where $\bar{E} = E/(1 - \nu^2)$ is the plane-strain Young’s modulus. Using the Gecit model³⁴, we calculate the normalized stress intensity factor $\hat{k}_n = K_h/K_o$ for a putative mode I tensile crack of normalized length $\hat{c} = c/h$ in ITO/PEN, ITO/PET, and ITO/PI film/substrate bilayer combinations specifically, whose Dundur’s parameters are listed in Supplementary Table 2 and plotted in Fig. 4b (see Supplementary Note 2 for modeling details^{34,39}). An incipient crack (small $\hat{c}$) at the surface of the film loaded in uniaxial tension (as in the bending test) will propagate when $K_h \geq K_{IC}$ condition is satisfied, where K_{IC} is the fracture toughness of the film. In such a loading situation, $K_o \propto c^{0.5}$, and, thus, the crack is expected to propagate unstably⁴⁰. The results in Fig. 4a show that, due to the presence of the compliant substrate, K_h for a crack in the film increases with $\hat{c}$, beyond the conventional $c^{0.5}$ -scaling increase represented by $\hat{k}_n = 1$. The $\hat{k}_n$ then rises dramatically as the crack approaches the interface ($\hat{c} \rightarrow 1$), and induces a stress intensity factor in the substrate at the interface that approaches infinity. The singularity implies that, in principle, the crack will continue to propagate into the substrate regardless of the substrate toughness, as hypothesized by Gecit³⁴ and Beuth^{35,36}. Fig. 4d–e schematically depict this substrate-cracking behavior. By corollary, in the absence of the film, the substrate would otherwise not crack due to its very high fracture toughness (Fig. 4c). Once the crack enters the substrate ($\hat{c} > 1$), $\hat{k}_n$ is expected to decrease as the crack propagates deeper into the substrate, an effect modeled by Thouless, et al.³⁷. In the case of bending, the applied σ_A experienced by the crack tip also decreases as it propagates deeper into the substrate, and the stress goes to zero at the neutral axis, before becoming compressive¹³. Thus, cracks in the substrate are limited to $\sim 20\%$ of its thickness, but are sufficiently deep to cause cracking-induced degradation of the device.

Fig. 4 | Modeling and mechanisms. **a** Plot of $\hat{k}_n$ ($= K_n/K_0$) as function of $\hat{c}$ ($= c/h$) for different film/substrate combinations relevant to this study. Inset: schematic illustration (not to scale) of the crack of depth, c , in the film of thickness, h . **b** Design map of the two Dundur's parameters (α and β) for film/substrate showing the boundary ($\alpha = -\beta$) between substrate-cracking ($\alpha > -\beta$) and crack-arrest ($\alpha < -\beta$) regions. The film/substrate combinations in (a) are indicated on the map in (b). **c–f** Schematic illustrations (not to scale) of cracking behavior under uniaxial tensile loading depicting no substrate cracking (c), 'channel' crack in film (d), extension of crack into the substrate (e), and substrate-cracking mitigation approach using an interlayer (f).



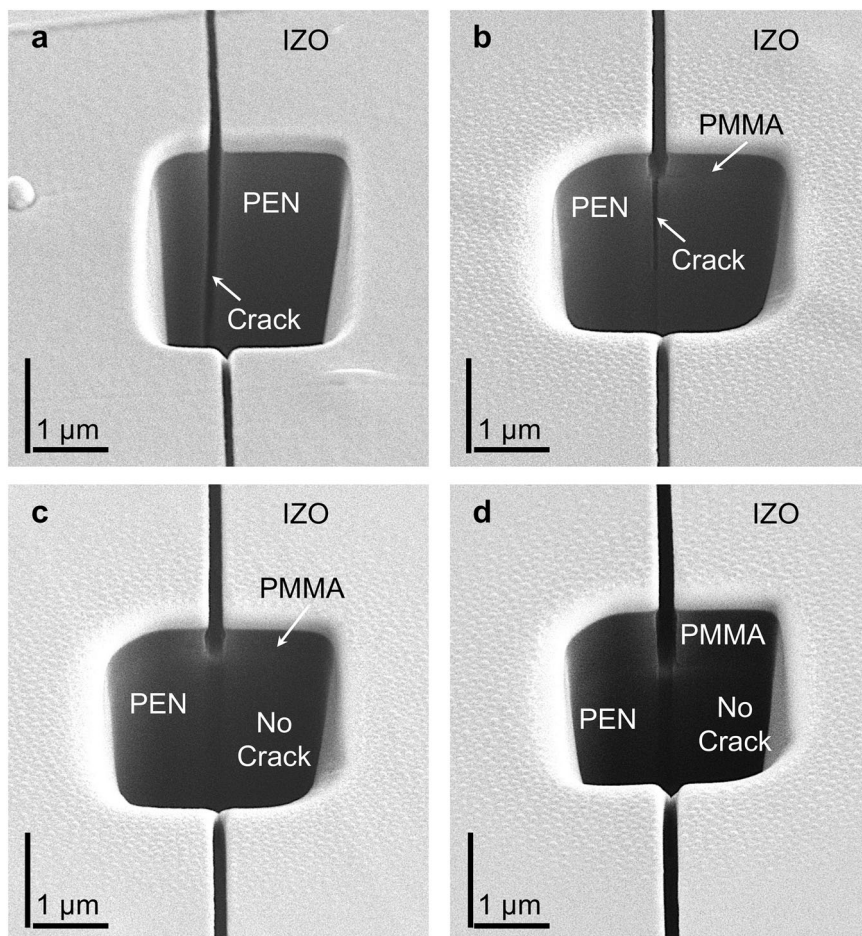
Substrate-cracking mitigation

Using the Gecit model³⁴, we have constructed a design map in Fig. 4b, where for $\alpha > -\beta$, $\hat{k}_n$ is expected to increase dramatically as the crack approaches the interface ($\hat{c} \rightarrow 1$), resulting in substrate cracking. By corollary, $\hat{k}_n$ will decrease with $\hat{c} \rightarrow 1$ for $\alpha < -\beta$, resulting in crack arrest at the interface. (Note that this map can also be derived from Zak-Williams analysis³³). As expected, ITO/PEN, ITO/PET, and ITO/PI combinations lie in the substrate-cracking region on the map in Fig. 4b. (IZO included in this map is another popular TCO that is used in this study, and its elastic properties are very similar to those of ITO (see Supplementary Table 1).) Since the much stiffer ITO (or IZO) film is key to the substrate-cracking problem, one can envision replacing the TCO with alternate, less stiff transparent-conducting electrodes made of polymers, metal meshes, or carbon-based nanomaterials⁴¹. However, those alternate electrodes are typically inferior to TCO in terms of the combination of high optical transparency and low sheet resistance they can offer⁴¹.

Using the design map in Fig. 4b, which provides general guidelines, we have designed a simple approach here for mitigating the substrate-cracking problem in generic TCO/polymer sheets for flexible electronic devices, without the need to eliminate the TCO. Our approach entails the use of a transparent, compliant interlayer between the TCO and polymer substrate, such that $\alpha < -\beta$ for the interlayer/substrate combination. To demonstrate the proof-of-concept, a polymethyl methacrylate (PMMA) interlayer ($E \sim 3$ GPa¹⁹) is chosen for insertion between IZO film ($E \sim 130$ GPa²²) and PEN substrate ($E \sim 9$ GPa¹⁵) (Supplementary Table 1). Figure 4a plots $\hat{k}_n$ as a function of $\hat{c}$ for the IZO/PMMA and PMMA/PEN combinations, whose Dundur's parameters are listed in Supplementary Table 2 and plotted in Fig. 4b. For the IZO/PMMA case, a very rapid rise in $\hat{k}_n$ is observed as the crack approaches the interface, commensurate with the large α and β for that

combination. Thus, a putative crack in IZO film is expected to propagate into the PMMA interlayer underneath, as depicted schematically in Fig. 4f. However, in the case of the PMMA/PEN combination, for the same putative crack in the PMMA layer, $\hat{k}_n$ diminishes as the crack approaches the PMMA/PEN interface. This is due to the reversal of the concentrated shear stresses induced in the substrate due to the elastic mismatch (Supplementary Fig. 3b). Therefore, the crack is expected to be arrested, precluding any substrate cracking (Fig. 4f). While the approach of introducing interlayers between dissimilar materials for mitigating cracking has been discussed before (see e.g.^{42,43}), this approach has eluded application to TCO/polymer sheets and related flexible electronic devices. To test this hypothesis experimentally, PMMA layers of different thicknesses (~ 250 , ~ 300 , ~ 400 , and ~ 600 nm) were spin-coated onto PEN substrates. Unfortunately, PMMA is unable to withstand the processing conditions of the subsequent IZO deposition. Therefore, an ultrathin, transparent layer of amorphous SiO₂ (~ 5 nm) was e-beam evaporated on the PMMA surface to protect it. Considering the thinness of the SiO₂, it is not expected to interfere with this mitigation approach. IZO (~ 250 nm thickness) was then sputter-deposited on top of the SiO₂. In the case of no-PMMA, substrate cracking is observed in the FIB-cut cross-section in Fig. 5a, which is similar to that in the ITO/PEN case (Fig. 1d). Similarly, in the case of 250-nm PMMA interlayer, cracking of the PEN substrate is observed in Fig. 5b. However, dramatic arrest of 'channel' cracks at the PMMA/PEN interface is observed in Fig. 5c with a ~ 300 nm PMMA interlayer. The existence of an interlayer critical thickness for effective substrate-crack mitigation is related to the length scale of the crack-tip stress field relative to the interlayer thickness. (This effect is not considered in the mechanics models discussed earlier.) Similar crack-arrest behavior is observed in Fig. 5d with a thicker PMMA interlayer (~ 600 nm). It is important to note that no evidence for interfacial

Fig. 5 | Mitigation of substrate cracking. a–d SEM images of FIB-cut cross-sections of ‘channel’ cracks in combinations: IZO/PEN (a), IZO/PMMA(250 nm)/PEN (b), IZO/PMMA(300 nm)/PEN (c), and IZO/PMMA(600 nm)/PEN (d), bent to $r = 7$ mm ($\epsilon_A = 0.0089$), in the bent state (strained). A thin SiO₂ layer (~ 5 nm) was used on top of PMMA. (FIB cuts were made while in the bent state. All FIB-cut specimens observed at a 52° forward tilt angle, hence vertical micron bars (representing depth) are longer than the horizontal ones, as indicated in (a–d)).



delamination was observed in Fig. 5a–d. (In related experiments, instead of the SiO₂ protective layer, Au layers (~ 10 nm) were deposited before and after PMMA deposition, which allowed better contrast delineation of the PMMA layer in the SEM; the same type of crack-arrest behavior is observed with a ~ 400 nm PMMA interlayer (Supplementary Fig. 4). This indicates that a relatively thin PMMA interlayer is sufficient for suppressing substrate cracking. Thus, the design map in Fig. 4b, in conjunction with static elastic properties of the materials under consideration, can be used as a general guide for choosing a suitable transparent material for interlayer between the TCO film and the substrate to prevent substrate cracking in those devices. Of course, the transparency requirement on the interlayer is relaxed for use in otherwise opaque flexible electronic devices. It is acknowledged that the incorporation of an additional layer (interlayer) between TCO/substrate can make the multilayer stack more susceptible to interfacial delamination, which is highly undesirable from a device-reliability standpoint. Thus, to avoid any interfacial delamination, it is important to ensure that the interfaces are sufficiently tough, as was done in this study.

Electrical properties

The relative change in the resistance ($\Delta R/R_0$) of the ITO film on PEN substrate was measured at different bending radii, r , in the bent state (strained), where the resistance R is in the direction perpendicular to the bending axis (Supplementary Fig. 5a), R_0 is the initial resistance in the flat state (unstrained), and $\Delta R = R - R_0$. Beyond a critical bending radius ($r = 10$ mm), ITO's $\Delta R/R_0$ increases monotonically with decreasing r . Supplementary Fig. 5b plots the corresponding measured density of ‘channel’ cracks as a function of decreasing r , in the bent state, which also increases monotonically. It is assumed that the increase in the crack density in the ITO film is responsible for the increase in the relative change in the

resistance in the bent state. However, when the ITO/PEN sheet is returned to the flat state (i.e. one bending cycle), $\Delta R/R_0$ typically approaches zero. This is because the mating walls of ‘channel’ cracks in the ITO can make good contact in the flat state, although the cracks are not healed.

In light of the substrate-cracking behavior reported here, it is hypothesized that when the TCO/polymer sheet is subjected to cyclic bending (i.e. multiple cycles), the propensity for the substrate cracks to close properly when returned to the flat state will decrease progressively. This is likely due to the accumulation of debris within the substrate cracks, worsening misalignment of the mating crack walls, and the fatigue-induced propagation of the crack deeper into the substrate, with increasing number of cycles. This, in turn, is likely to progressively prevent the TCO crack walls from making good contact, thereby resulting in increasing $\Delta R/R_0$ (flat state) with number of cycles, n . It is also hypothesized that in the case where substrate cracking is eliminated, e.g. in IZO/PMMA(300 nm)/PEN, the flat-state $\Delta R/R_0$ may not increase with increasing n . These hypotheses are illustrated schematically in Supplementary Fig. 6. To verify these hypotheses, $\Delta R/R_0$ of IZO/PEN sheets was measured in the flat state (Fig. 6) after subjecting them to bending cycles in the range $n = 1$ to 10^4 , at bending radius $r = 7$ mm ($\epsilon_A = 0.0089$) using a testing method described elsewhere²⁸. Fig. 6 plots these data, which clearly show progressive increase in flat-state $\Delta R/R_0$ with number of bending cycles, from ~ 0.1 to $\sim 1,200$ after 10^4 bending cycles. In contrast, no appreciable change in $\Delta R/R_0$ could be measured in IZO/PMMA(~ 300 nm)/PEN in the flat state, even after 10^4 cycles where $\Delta R/R_0 \sim 0.5$. These results reinforce the critical importance of substrate cracking in the degradation of TCO electrical properties. Since electrical continuity of the TCO film is key to the proper functioning of any multilayer device built upon TCO/polymer sheets, substrate cracking is highly detrimental for the durability and reliability of such devices.

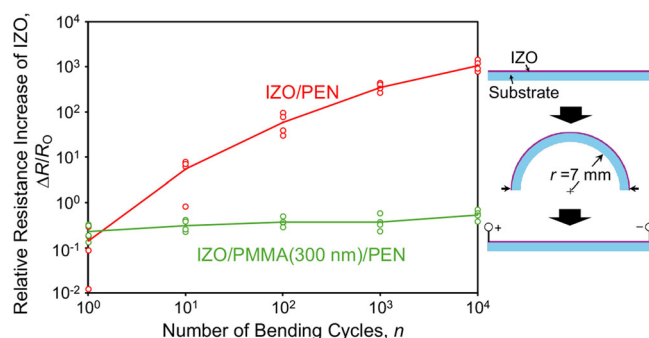


Fig. 6 | Cyclic bending and electrical properties. The relative increase in the DC electrical resistance of IZO film on PEN substrate, where the substrate cracks in bending ($r = 7$ mm; $\epsilon_A = 0.0089$), in the flat state (unstrained) as a function of bending cycles, n . Corresponding measurements of IZO on PMMA(300 nm)/PEN substrate, where the substrate does not crack in bending. Measurements from four different specimens for each material system reported. The solid lines connect the averages. The schematic illustration (not to scale) on the right depicts one bending cycle: flat $\rightarrow$ bent $\rightarrow$ flat.

Discussion

While the severity and extent of polymer-substrate cracking in TCO/polymer sheets will depend on the film/substrate materials, their relative thicknesses, and applied tensile stresses, the substrate-cracking mitigation approach of introducing a more compliant interlayer above the substrate is potentially applicable to myriad flexible electronic devices. It is imperative that consideration of polymer substrate cracking, and its mitigation, become integral part of design, manufacturing, and operation of future flexible electronic devices with improved durability and reliability.

Methods

Materials

All reagents/materials were used as-received commercially without further purification, which include: lead(II) iodide (PbI_2 ; 99%, Sigma Aldrich, USA), formamidine iodide (FAI; >99.99%, Greatcell Solar Materials, Australia), lead(II) bromide (PbBr_2 ; Materials, Australia), cesium iodide (CsI ; 99.999%, Sigma Aldrich, USA), methylammonium bromide (MABr; >99.99%, Greatcell Solar Materials, Australia), methylammonium chloride (MACl; >99.99%, Greatcell Solar Materials, Australia), N,N-dimethylformamide (DMF; 99.8%, anhydrous, Sigma Aldrich, USA), dimethyl sulfoxide (DMSO; $\geq 99.9\%$, anhydrous, Sigma Aldrich, USA), ethanol (>99.5%; Sigma Aldrich, USA), acetone (>99.5%, VWR, USA), SnO_2 colloidal dispersion (SnO_2 -CD; 15 wt % in H_2O , Alfa Aesar, USA), isopropanol (IPA; 99.5%, anhydrous, Sigma Aldrich, USA), bis(trifluoromethane)sulfonimide lithium salt (Li-TFSI; 99.95%, Sigma Aldrich, USA), 4-tert-butylpyridine (tBP; 98%, Sigma Aldrich, USA), acetonitrile (AcN; 99.8%, anhydrous, Sigma Aldrich, USA), chlorobenzene (CB; 99.8%, anhydrous, Sigma Aldrich, USA), spiro-OMeTAD (>99.8%, Lumtec, Taiwan), polymethylmethacrylate solution (PMMA; Sigma Aldrich (182265), USA), polyethylene naphthalate sheets (PEN; Goodfellow Corp., USA), polyimide sheets (PI; Kapton, Kolon Industried, S. Korea), indium-tin-oxide-coated polyethylene terephthalate substrates (ITO/PET; Sigma Aldrich, USA), and ITO-coated PEN substrates (ITO/PEN; Peccell, Japan).

Deposition of PMMA thin films

Solution of PMMA in CB (5 wt%, heated at 70 °C overnight) was spin-coated on PEN substrates under ambient conditions at 6000, 4000, 2000, or 1000 rpm, each with a ramp of 500 rpm s^{-1} , for 30 s to achieve a thicknesses of ~250, ~300, ~400, or ~600 nm, respectively.

Deposition of SiO_2 and Au Thin Films

SiO_2 thin films of ~5 nm thickness were deposited on top of the PMMA thin films at room temperature using an electron-beam evaporator (Ångström,

USA), at a rate of 1 Å s^{-1} . In related experiments, instead of SiO_2 thin films, Au thin films (~10 nm) were deposited by e-beam evaporation (Ångström, USA).

Deposition of ITO thin films

ITO thin films (~200 nm thickness) were deposited on PI substrates (~80 μm thickness) using a radio frequency (RF) linear sputtering system (Kenosistec, Italy). The chamber was first evacuated to 5×10^{-6} Torr, followed by a pre-sputtering phase to cleanse the target and enhance the process reproducibility. Deposition occurred under 1.1×10^{-3} Torr pressure with an Ar flow of 40 SCCM and a power density of 0.39 W cm^{-2} . During the deposition, the substrate was moved laterally to ensure uniform deposition.

Deposition of IZO thin films

IZO thin films (~250 nm thickness) were deposited on the SiO_2 /PMMA/PEN or Au/PMMA/Au/PEN sheets using DC-magnetron sputtering at room temperature using an IZO ceramic target (90 wt.% In_2O_3 -10 wt.% ZnO). Specimens were attached to a metal mask and placed in the chamber, which was then evacuated until the pressure dropped below 5×10^{-6} Torr. Ar gas was passed through the gas mass flow controller into the chamber at 8 SCCM. The power supply for the plasma generator was then turned on and the working pressure was varied such that the DC bias was 280-300 V. After comparing different deposition times and measuring the respective film thickness, the deposition rate was determined to be ~0.25 Å s^{-1} .

Flexible PSCs fabrication

Flexible PSCs were fabricated using a procedure described elsewhere⁴⁴. Briefly, the ITO/PEN sheets were patterned using the Zn-HCl chemical-etching method. They were then sonicated in a solution of IPA, acetone, and DI (1:1:1 v/v) for 20 min. The substrates were subsequently dried under flowing dry N_2 and attached to an appropriately-sized microscope glass slide using tape. The substrates were treated under ultraviolet ozone (UVO) for 30–35 min and transferred to a controlled-humidity (<10% RH) glovebox. Immediately after this, an SnO_2 solution (SnO_2 -CD in DI water; 1:3 v/v, filtered with a 0.22 μm PTFE filter) was spin-coated at 4,000 rpm with a ramp of 1,000 rpm s^{-1} for 30 s, followed by drying at 120 °C for 60 min on a hotplate. The metal-halide perovskite (MHP) precursor solution (1.61 M) was prepared by dissolving 742.2 mg of PbI_2 , 224.4 mg of FAI, 16.2 mg of MABr, 20.3 mg of MACl, and 19.8 mg of CsI in a mixture of 800 μL DMF and 200 μL DMSO. This results in thin films of final nominal composition of $\text{Cs}_{0.05}\text{MA}_{0.13}\text{FA}_{0.82}\text{Pb}(\text{I}_{2.9}\text{Br}_{0.01})$. This precursor solution was stirred for 2 h at room temperature and filtered before use. After treating the SnO_2 -coated substrates under UVO for 15 min, the MHP layer was spin-coated on the prepared substrate at 1,000 rpm, with a ramp of 200 rpm s^{-1} for 10 s followed by 5000 rpm, with a ramp of 1000 rpm s^{-1} , for 30 s. Anhydrous CB (0.2 mL) was quickly dripped in a single stream onto the rotating substrate 15 s before the end. Subsequently, the film was annealed for 30–40 min at 100 °C under ~17% RH on a hotplate. The humidity was then reduced to <10% RH. The Spiro-OMeTAD layer was then spin-coated at 4,000 rpm, with a ramp of 2,000 rpm s^{-1} , for 20 s using a solution of 90 mg Spiro-OMeTAD, 39.5 μL of tBP, 23 μL of Li-TFSI (520 mg dissolved in 1 mL of acetonitrile) in 1 mL CB (filtered with 0.22 μm PTFE filter). All processing was performed in a humidity-controlled ambient-air glovebox. The flexible PSCs were then detached from the glass slide. Finally, 70-nm thick Au electrode was deposited by thermal evaporation.

Bending tests and FIB cutting

Focused ion beam (FIB) cutting of cross-sections was performed using a Helios 600 instrument (ThermoFisher, USA) with all the specimens being in the bent state. The different flexible specimens were carefully wrapped around custom-made acrylic cylindrical mandrels with r ranging from 5 to 10 mm. In each case, the length of the substrate was $\sim \pi r$. After sputtering a thin layer of conductive carbon, the mandrels were carefully attached to the FIB stage using carbon tape. Additional copper tapes were used to hold the sample down, as well as to reduce any charging effects. A beam current of

0.46–0.92 nA was used to make the FIB cuts, in cleaning cross-section mode, without any Pt deposition. It took ~3–5 min to make each FIB cut. In some experiments, post-bending etching of the ITO film was performed using the Zn-HCl method.

Characterization

All scanning electron microscope (SEM; Quattro S, ThermoFisher, USA) observations, top view and cross-sectional view, were performed while the specimens were in the bent state. The same procedure as above was used to mount the bent specimens onto the SEM stage.

An optical microscope (Eclipse LV100, Nikon, USA) was used to make top-view observations, and also to measure the crack densities while the specimens were in the bent state. The same procedure as above was used to mount the bent specimens onto the optical microscope stage. The number of cracks in an image was divided by the corrected image length, to account for the curvature, and this number is referred to as the crack density. For each r condition, a minimum of three specimens were used for this measurement.

Electrical properties measurement

A multimeter (179, Fluke, USA) was used to measure the DC electrical resistance of the ITO film in ITO/PEN. Copper tapes were attached to the ends of the ITO film, and the initial resistance (R_0) in the flat state was carefully measured, which was found to be very close to a value calculated from its specified sheet resistance of the ITO film (13–15 $\Omega/\square$). The ITO/PEN specimens were manually bent on the acrylic cylindrical mandrels of different r (5, 7, or 10 mm). Copper tapes were attached to the flat part of the ITO film at either ends, and they were then connected to the multimeter to measure the resistance (R) of the ITO film in the bent state. For each r condition, about four specimens were used for this measurement. All bending and measurement procedures were performed in a dry-air glovebox (<5% RH).

Cyclic bending tests were performed on IZO/PEN and IZO/PMMA(300 nm)/PEN sheets in the dry-air glovebox (RH<5%) using an automatic cyclic mechanical tester (PR-BDM-100, Puri, China), where the acrylic cylindrical mandrel was of radius $r = 7$ mm; each bending cycle is: flat→bent→flat²⁸. Testing was performed for 10,000 cycles, at the rate of ~1 cycle·s⁻¹. Testing was interrupted periodically, and the DC electrical resistance of the IZO films in the flat state were measured using the method described above. The initial resistance (R_0) values of the IZO films in the flat state were measured before the commencement of cyclic bending tests.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

N.P.P. conceived and supervised the research. N.P.P. and A.R. designed the research. A.R. performed bulk of the experimental work, including synthesis, processing, fabrication, characterization, measurements and testing, with contributions from M.L., C.L-P., A.C., C.K., E.M., A.D.C., F.B., P.G., and D.P. The mechanics modeling work and the interpretation of the results was performed by S.K. and H.K. The data were interpreted by N.P.P., A.R., H.K., and S.S. The manuscript was written by N.P.P. and A.R., with contributions from the other authors.

Competing interests

The authors declare no competing interests.

Additional information

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